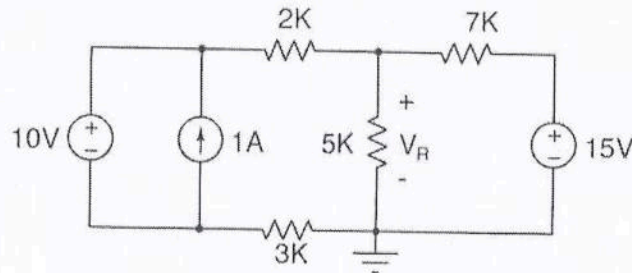


- SOLUTIONS -

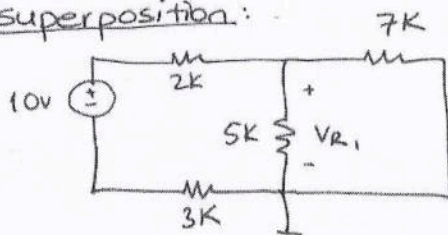
Question 1 [10 Points]:

In the following circuit, find V_R with any method of your choice.



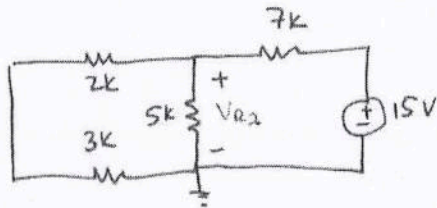
Many ways to solve. Here are two options:

a) Superposition:



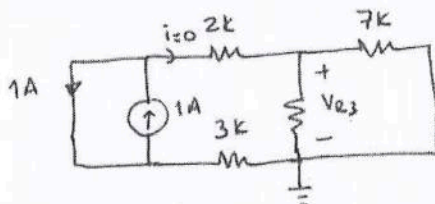
$$V_{R1} = 10 \cdot \frac{5K // 7K}{2K + 3K + 5K // 7K} \quad (\text{voltage divider})$$

$$= 10 \cdot \frac{\frac{35}{12}}{5 + \frac{35}{12}} = 10 \cdot \frac{35}{95} = \frac{70}{19} V$$



$$V_{R2} = 15 \cdot \frac{(2K + 3K) // 5K}{7K + (2K + 3K) // 5K} \quad (\text{voltage divider})$$

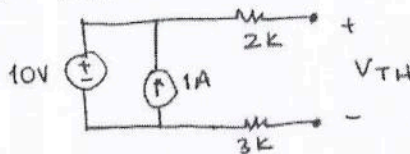
$$= 15 \cdot \frac{5/2}{7 + 5/2} = \frac{75}{19} V$$



$$V_{R3} = 0 V \quad (\text{due to short circuit})$$

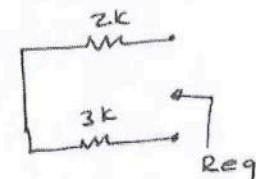
$$V_R = V_{R1} + V_{R2} + V_{R3} = \frac{145}{19} V = \boxed{7.63 V}$$

b) Thevenin Eq.

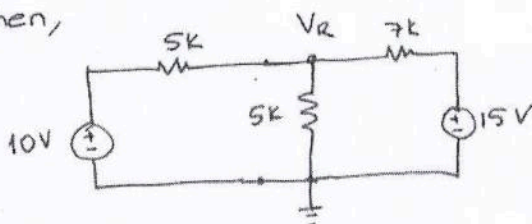


$$V_{TH} = 10V$$

$$R_{eq} = 5K\Omega$$



Then,



$$\frac{V_R - 10}{5K} + \frac{V_R}{5K} + \frac{V_R - 15}{7K} = 0$$

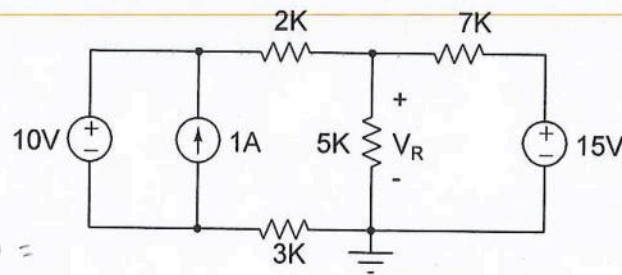
$$7V_R - 70 + 7V_R + 5V_R - 75 = 0$$

$$19V_R = 145$$

$$V_R = \frac{145}{19} V = \boxed{7.63 V}$$

Question 1 [10 Points]:

In the following circuit, find V_R with any method of your choice.



$$5/7 = 2.9166$$

$$5k/15k = 2.5k$$

By superposition =

$$V_R = 10 \times \frac{2.9166}{2+3+2.9166} + 15 \times \frac{2.5}{2.5+7}$$

$$= 3.6842 + 3.947 = 7.6315 \text{ V}$$

Keep in mind that the current source does not have any effect on the circuit rather than pumping current into the 10V voltage source.

Q. 1 Solution by nodal analysis

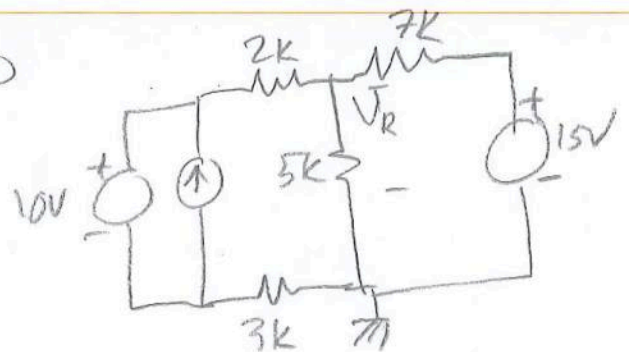
$$\frac{V_R}{5k} + \frac{V_R - 10}{5k} - \frac{V_R - 15}{7k} = 0$$

$\times (7) \quad \times (7) \quad \times (5)$

$$7V_R + 7V_R - 70 + 5V_R - 75 = 0$$

$$19V_R - 145 = 0 \Rightarrow$$

$$V_R = \frac{145}{19} = 7.6315$$



Typical wrong solution by nodal analysis

$$\frac{V_R}{5k} + \frac{V_R - 10}{2k} + \frac{V_R - 15}{7k} = 0$$

$\times (14) \quad \times (35) \quad \times (10)$

$$14V_R + 35V_R - 350 + 10V_R - 150 = 0$$

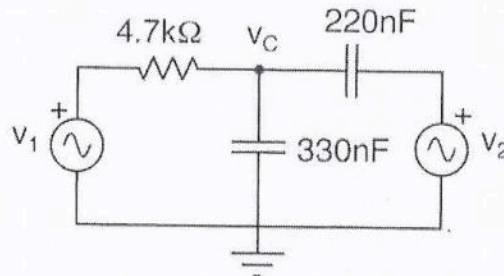
$$59V_R - 500 = 0 \Rightarrow$$

$$V_R = \frac{500}{59} = 8.47$$

Question 2 [15 Points]:

Find the voltage across the 330nF capacitor, $v_C(t)$, if the voltage sources are

$$v_1(t) = 12 \cos(1000t) \text{ Volts, and } v_2(t) = 7 \cos(3000t + 45^\circ) \text{ Volts.}$$

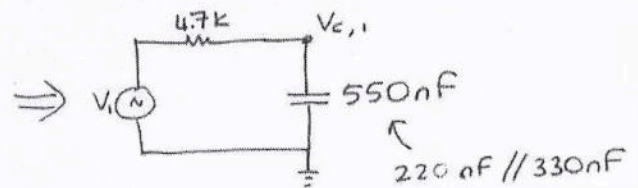
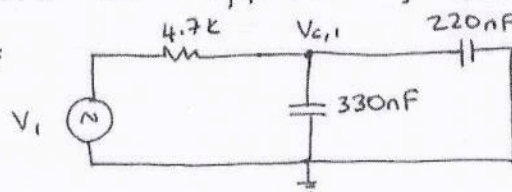


$$C_1 = 330 \text{ nF}$$

$$C_2 = 220 \text{ nF}$$

Inputs at different frequencies. Use superposition.

From V_1 :



$$V_1 = 12 \text{ V (in phasor domain)}$$

$$\frac{1}{j\omega_1 C_T} = \frac{1}{j1000 \cdot 550 \cdot 10^{-9}} = -j1.82 \text{ k}\Omega$$

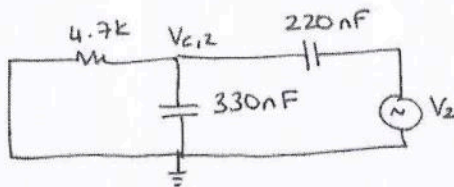
using calculator

$$V_{C,1} = V_1 \cdot \frac{-j1.82 \text{ k}}{4.7 \text{ k} - j1.82 \text{ k}} = 12 \times 0.3611 \cdot e^{-j1.2} = 4.33 \cdot e^{-j1.2}$$

$$v_{C,1}(t) = 4.33 \cos(1000t - 1.2) \text{ V} = 4.33 \cos(1000t - 68.8^\circ) \text{ V}$$

in radians in degrees

From V_2 :



$$V_2 = 7 \cdot e^{j\pi/4}$$

$$\frac{1}{j\omega_2 C_1} = \frac{1}{j3000 \cdot 330 \cdot 10^{-9}} = -j1.01 \text{ k}\Omega$$

$$\frac{1}{j\omega_2 C_2} = \frac{1}{j3000 \cdot 220 \cdot 10^{-9}} = -j1.52 \text{ k}\Omega$$

$$V_{C,2} = V_2 \cdot \frac{4.7 \text{ k} \parallel (-j1.01 \text{ k})}{-j1.52 \text{ k} + 4.7 \text{ k} \parallel (-j1.01 \text{ k})} = 7 \cdot e^{j\pi/4} \cdot \frac{-j4.747}{-j1.52(4.7 - j1.01) - j4.747}$$

$$= 7 \cdot e^{j\pi/4} \cdot 0.3959 \cdot e^{j0.3959} = 2.77 \cdot e^{j0.91}$$

$$v_{C,2}(t) = 2.77 \cos(3000t + 0.91) \text{ V} = 2.77 \cos(3000t + 52.3^\circ) \text{ V}$$

in radians in degrees

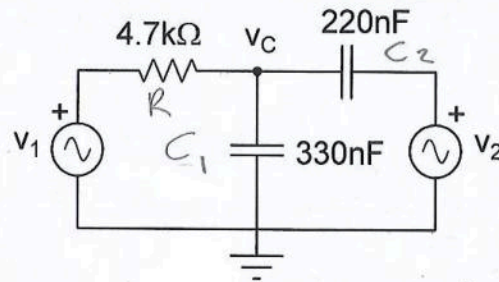
* Finally,

$$v_C(t) = v_{C,1}(t) + v_{C,2}(t)$$

Question 2 [15 Points]:

Find the voltage across the 330nF capacitor, $v_C(t)$, if the voltage sources are

$$v_1(t) = 12 \cos(1000t) \text{ Volts, and } v_2(t) = 7 \cos(3000t + 45^\circ) \text{ Volts.}$$



For $V_{Cv_1} =$ $330 + 220 = 550 \text{ nF}$ $|X_C| = \frac{1}{1000 \times 550 \times 10^{-9}} = 1818.18 \Omega$

$$\vec{V}_{Cv_1} = 12 \cdot \frac{-j1818.18}{4700 - j1818.18} = \frac{-j1818.18 \times 10^3}{4.7 - j1.818} \text{ V}$$

$$= 12 \times \frac{-j1.818(4.7 + j1.818)}{25.4} = \frac{21.816[4.7 + j1.818](-j)}{25.4}$$

$$\vec{V}_{Cv_1} = \frac{21.816 \times 25.4^{1/2}}{25.4} \angle 90 - 21.147^\circ \text{ at } \omega = 1000 \text{ rad/s/sec}$$

$$= 4.3287 \angle -68.85^\circ$$

For V_{Cv_2} $|X_{C1}| = \frac{1}{3000 \times 0.33 \times 10^{-6}} = \frac{1}{10^3} = 1010 \Omega$

$$|X_{C2}| = \frac{1}{3000 \times 0.22 \times 10^{-6}} = \frac{1}{0.66 \times 10^{-3}} = 1.515 \times 10^3 = 1515 \Omega$$

$$R \parallel X_{C1} = \frac{4700 \times (-j1010)}{4700 - j1010} = 10^3 \cdot \frac{-4.78105}{4.7 - j1.01} = 10^3 \cdot \frac{-4.78105(4.7 + j1.01)}{(4.7^2 + 1.01^2)}$$

$$\frac{4.747 - j22.09}{23.11} \times 10^3 = 10^3 \cdot \frac{2.2.59}{23.11} \angle -77.87^\circ = 977.7 \angle -77.87^\circ$$

$$\vec{V}_{C2} = 7 \angle 45^\circ \cdot \frac{977.7 \angle -77.87^\circ}{205.4 - j955.9 - j1515} = 7 \angle 45^\circ \times \frac{977.7 \angle -77.87^\circ}{205.4 - j2470.87}$$

$$= 7 \angle 45^\circ \cdot \frac{977.7 \angle -77.87^\circ}{2479.3 \angle -85.26^\circ} = 2.76 \angle 52.27^\circ$$

$$v_C = 4.33 \cos[1000t - 68.85^\circ] + 2.76 \cos(3000t + 52.27^\circ)$$

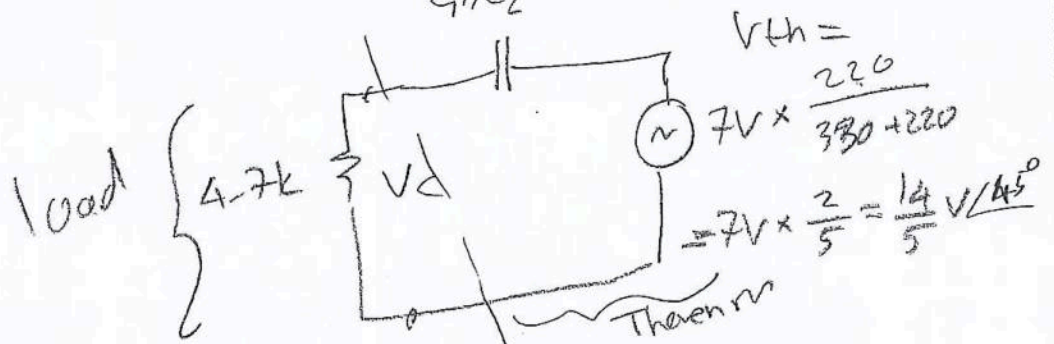
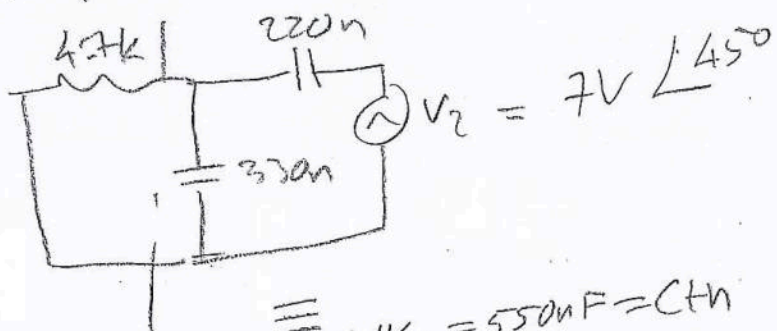
Alternative solution for Q.2 for V_1

$$V_1 = 12 \times \frac{\frac{1}{j\omega C_T}}{4700 + \frac{1}{j\omega C_T}} = \frac{1}{1 + 4700j\omega 550 \times 10^{-9}} = \frac{12}{1 + 2.585j}$$

$$= \frac{12}{(1^2 + 2.585^2)^{1/2}} \angle -\arctan 2.585 = \frac{12}{2.777} \angle -68.85^\circ$$

$$V_1 = 4.33 \angle -68.85^\circ$$

Alternative solution for Q.2 for V_2



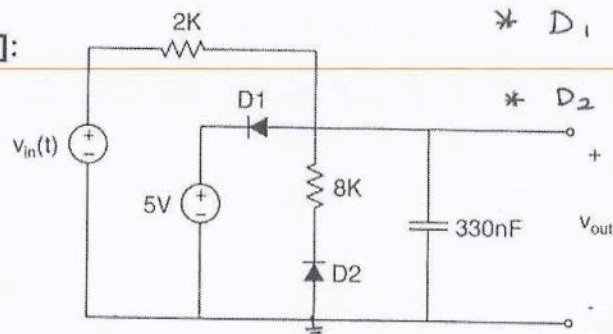
$$V_c = V_{th} \cdot \frac{4.7k}{4.7k + \frac{1}{j\omega C_{TH}}} = \frac{14}{5} \angle 45^\circ \cdot \frac{4700}{4700 - j606.1}$$

$$= \frac{14}{5} \angle 45^\circ \cdot \frac{4.7}{4.7 - j606.1} = \frac{14}{5} \angle 45^\circ \cdot \frac{1}{1 - 0.1289j}$$

$$= 2.8 \angle 45^\circ \times \frac{1}{1 - 0.1289j} = 2.8 \angle 45^\circ \cdot \frac{1}{1.0083} \angle 7.45^\circ$$

$$= 2.777 \angle 52.3^\circ$$

Question 3 [25 Points]:

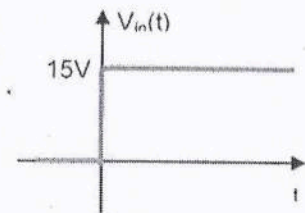


* D_1 turns ON when V_{out} reaches 6V

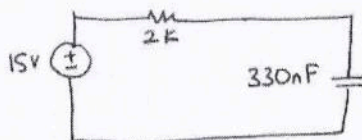
* D_2 turns ON when V_{out} reaches -1V

In the circuit above, the threshold voltage $V_{ON} = 1V$ for the diodes D_1 and D_2 , and $V_C(0) = 0V$ for the capacitor.

- (a) [7 points] Find and plot $V_{out}(t)$ for $t > 0$ if $V_{in}(t)$ is as shown on the right. Mark all important voltage levels and time points. Clearly specify the states of the diodes for $t > 0$. D_2 cannot turn ON.



* Initially, both diodes are off. So,



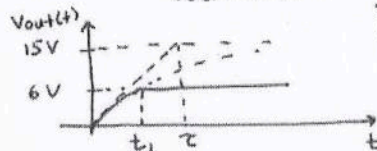
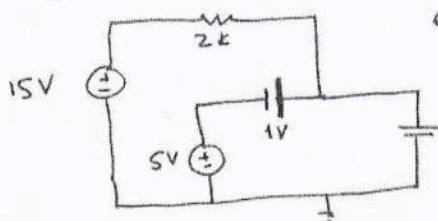
$$V_{init} = 0$$

$$V_{final} = 15V$$

$$\tau_1 = 2k\Omega \cdot 330nF = 0.66ms$$

$$V_{out}(t) = V_{final} + (V_{init} - V_{final})e^{-t/\tau_1} = 15 - 15e^{-t/\tau_1} V$$

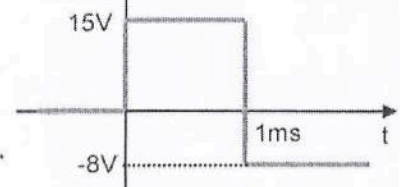
* D_1 turns ON when V_{out} reaches $5V + V_{ON} = 6V$. After that: capacitor in parallel with 6V voltage source. Charging stops. This happens when:



$$V_{out}(t_1) = 15 - 15e^{-t_1/\tau_1} = 6V \Rightarrow t_1 = 0.34ms$$

$$V_{out}(t) = \begin{cases} 15 - 15e^{-t/\tau_1} V, & 0 < t < t_1 \\ 6V, & t \geq t_1 \end{cases}$$

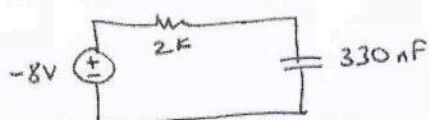
- (b) [18 points] Find and plot $V_{out}(t)$ for $t > 0$ if $V_{in}(t)$ is as shown on the right. Mark all important voltage levels and time points. Clearly specify the states of the diodes for $t > 0$.



at $t = 1ms$, $V_{init} = 6V$ (from part (a))

D_1 turns OFF, and cannot turn on again.

* D_2 initially OFF. So,

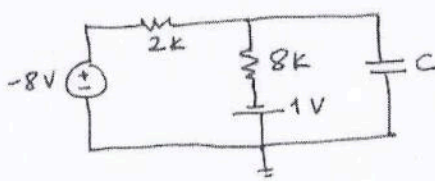


$$V_{final} = -8V, \tau_1 = 2k\Omega \cdot 330nF = 0.66ms$$

$$V_{out}(t) = -8 + (6 - (-8))e^{-(t-1)/\tau_1} = -8 + 14e^{-(t-1)/\tau_1} V$$

* D_2 turns ON when V_{out} reaches -1V. This happens when $V_{out}(t_2) = -8 + 14e^{-(t_2-1)/\tau_1} = -1V \Rightarrow t_2 = 1.46ms$

* when D_2 turns ON,



Now, replace C with open circuit for V_{final} : $\frac{V_f - (-8V)}{2k} + \frac{V_f - (-1V)}{8k} = 0 \Rightarrow V_{final} = -6.6V$

$$\tau_2 = (2k\Omega // 8k\Omega) \cdot 330nF = 1.6k\Omega \cdot 330nF = 0.53ms$$

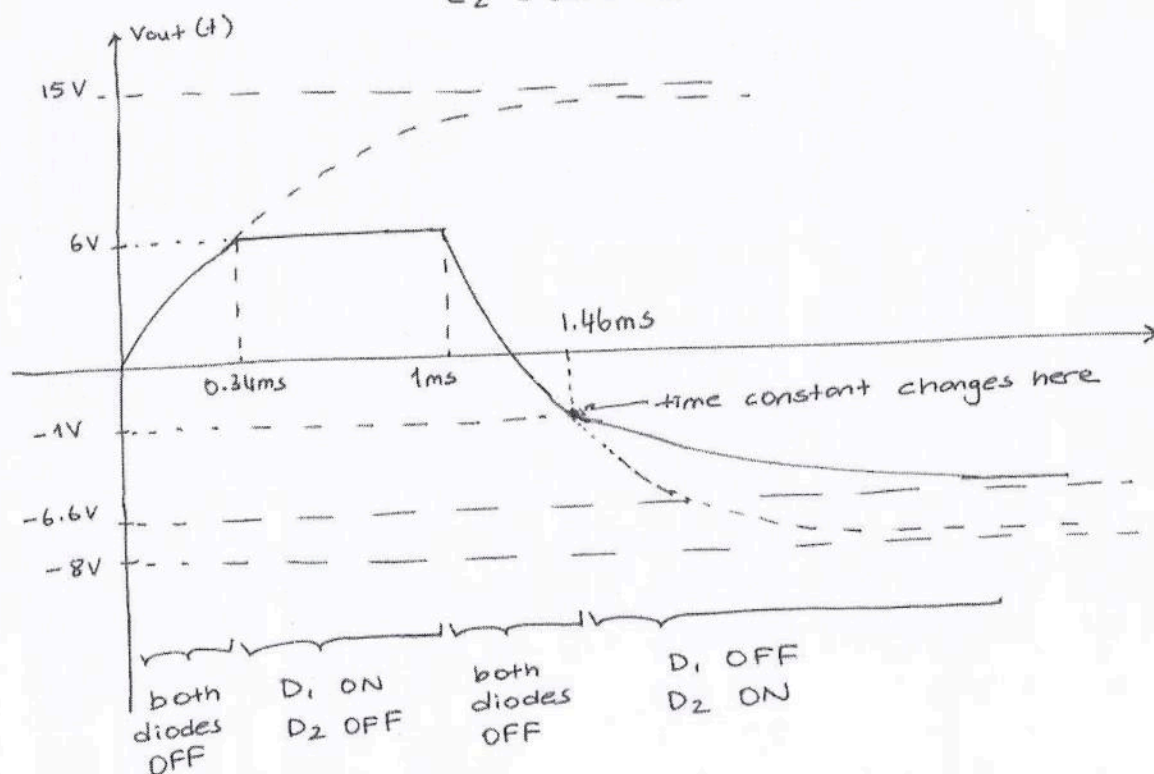
$$V_{out}(t) = -6.6 + (-1 - (-6.6))e^{-(t-1.46)/\tau_2}$$

$$= -6.6 + 5.6e^{-(t-1.46)/\tau_2} V$$

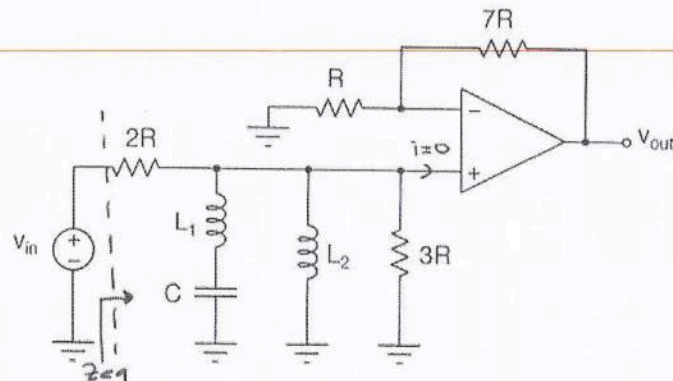
So,

$$V_{out}(t) = \begin{cases} 15 - 15 e^{-t/\tau_1} \text{ V} & , 0 < t < 0.34 \text{ ms} \\ 6 \text{ V} & , 0.34 \text{ ms} \leq t < 1 \text{ ms} \\ -8 + 14 e^{-(t-1)/\tau_1} \text{ V} & , 1 \text{ ms} \leq t < 1.46 \text{ ms} \\ -6.6 + 5.6 e^{-(t-1.46)/\tau_2} \text{ V} & , 1.46 \text{ ms} \leq t \end{cases}$$

where $\tau_1 = 0.66 \text{ ms}$, and t in ms
 $\tau_2 = 0.53 \text{ ms}$



Question 4 [25 points]:



In the circuit above, $L_1 = 1\text{mH}$, $L_2 = 2\text{mH}$, $C = 3\text{nF}$, $R = 10\text{k}\Omega$. Assume that the OPAMP is not saturated and is an ideal OPAMP.

(a) [8 points] What is the resonance frequency of this circuit in units of Hz?

$$Z_{eq} = 2R + \left(\frac{1}{3R} + \frac{1}{j\omega L_2} + \frac{1}{j\omega L_1 + \frac{1}{j\omega C}} \right)^{-1} = 2R + \left(\frac{1}{3R} + \frac{1}{j\omega L_2} + \frac{j\omega C}{1 - \omega^2 L_1 C} \right)^{-1}$$

$$= 2R + \left(\frac{1}{3R} + j \left[\frac{\omega^2 L_2 C + \omega^2 L_1 C - 1}{\omega L_2 (1 - \omega^2 L_1 C)} \right] \right)^{-1}$$

at resonance, $\text{Im}\{Z_{eq}\} = 0$,

this term should be zero

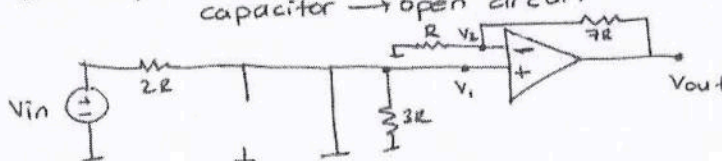
$$\text{So, } \omega^2 L_2 C + \omega^2 L_1 C - 1 = 0$$

$$\omega_0 = \frac{1}{\sqrt{(L_1 + L_2)C}}$$

$$f_0 = \frac{1}{2\pi\sqrt{(L_1 + L_2)C}} = \boxed{53.05 \text{ kHz}}$$

(b) [5 points] What is V_{out} if $V_{in} = 2\text{V}$ (DC)?

at DC, inductor $\rightarrow$ short circuit
capacitor $\rightarrow$ open circuit

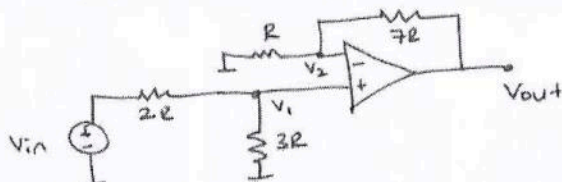


$$V_1 = V_2 = 0\text{V}$$

$$\text{Then, } \boxed{V_{out} = 0\text{V}}$$

(c) [5 points] Find $v_{out}(t)$ if $v_{in}(t) = 5 \cos(\omega_0 t + \pi/6)$ Volts, with ω_0 at resonance frequency.

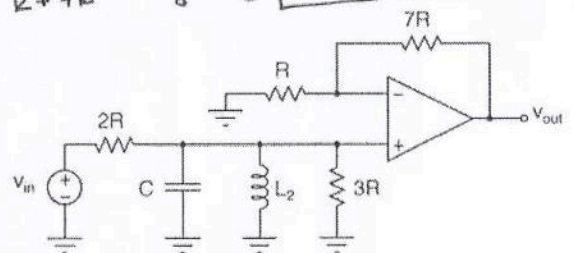
at resonance, inductors and capacitor together act like open circuit. (see Z_{eq} above)



$$\left. \begin{aligned} V_1 &= V_{in} \cdot \frac{3R}{2R + 3R} = V_{in} \cdot \frac{3}{5} \\ V_2 &= V_{out} \cdot \frac{R}{R + 7R} = \frac{V_{out}}{8} \end{aligned} \right\} \begin{aligned} V_1 &= V_2 \\ V_{out} &= V_{in} \cdot \frac{2.4}{5} \\ \boxed{V_{out} &= 25 \cos(\omega_0 t + \pi/6) \text{ V}} \end{aligned}$$

(d) [6 points] Now, L_1 is replaced with a short circuit as shown on the right.

- [2 pts] What is the new resonance frequency, f_0 ?
- [2 pts] What is the Q?
- [2 pts] What is the 3-dB bandwidth in Hz?



$$\text{i) } f_0 = \frac{1}{2\pi\sqrt{L_2 C}} = \boxed{64.98 \text{ kHz}}$$

$$\text{ii) } Q = \frac{2R // 3R}{\omega_0 L_2} = \frac{\frac{6}{5} \cdot 10 \cdot 10^3}{2\pi \cdot 64.98 \cdot 10^3 \cdot 2 \cdot 10^{-3}} = \boxed{14.7}$$

$$\text{iii) } \Delta f = \frac{f_0}{Q} = \frac{64.98 \text{ kHz}}{14.7} = \boxed{4.42 \text{ kHz}}$$

Question 4 (a) and (c): Alternative Solution.

(a) There is a second resonance frequency for this circuit.

$$Z_{eq} = 2R + \left(\frac{1}{3R} + j \left[\frac{\omega^2 L_2 C + \omega^2 L_1 C - 1}{\omega L_2 (1 - \omega^2 L_1 C)} \right] \right)^{-1}$$

For $\text{Im}\{Z\} = 0$, this part either should be zero (see previous page)

OR infinite.

i.e., either the denominator or the numerator should be zero.

Denominator zero: $1 - \omega^2 L_1 C = 0 \Rightarrow f_0 = \frac{1}{2\pi\sqrt{L_1 C}} = \boxed{91.89 \text{ kHz}}$

Here, L_1 & C are series resonant.

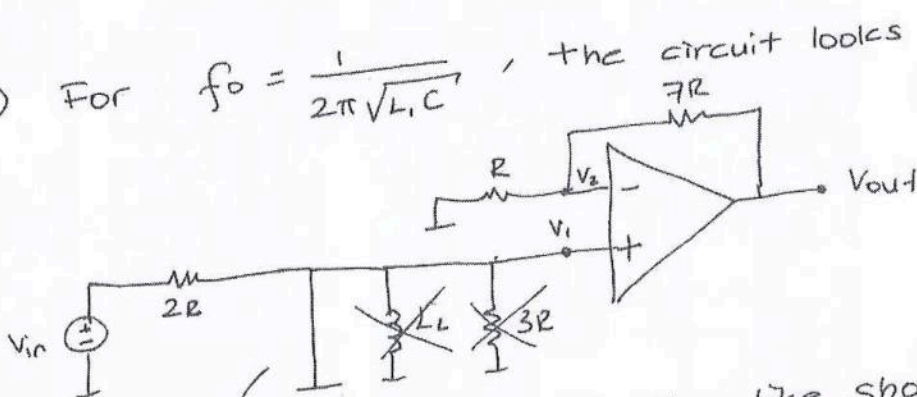
Together, they act like short circuit. $\Rightarrow Z_{eq} = 2R$
(see Z_{eq} above)

Numerator zero: $\omega^2 L_2 C + \omega^2 L_1 C - 1 = 0 \Rightarrow f_0 = \frac{1}{2\pi\sqrt{(L_1 + L_2)C}} = \boxed{53.05 \text{ kHz}}$

Here, L_1 & L_2 & C are resonant.

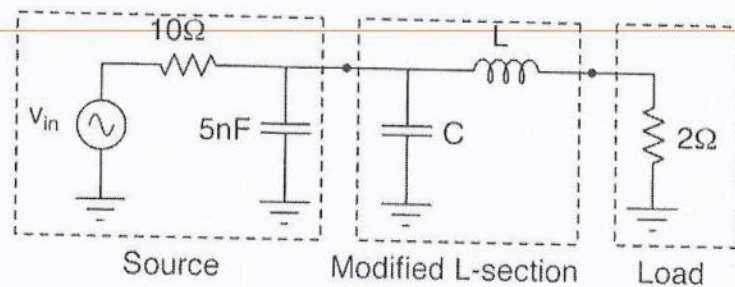
Together, they act like open circuit. $\Rightarrow Z_{eq} = 5R$
(see Z_{eq} above)

(c) For $f_0 = \frac{1}{2\pi\sqrt{L_1 C}}$, the circuit looks like



due to L_1 & C acting like short circuit,
 $V_1 = 0$, so, $\boxed{V_{out} = 0}$ at this resonance frequency.

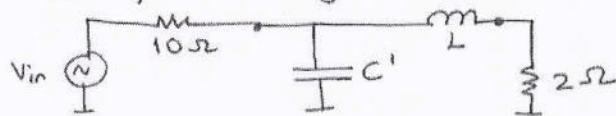
Question 5 [25 Points]:



In the circuit above, $v_{in}(t) = 5 \cos(10^7 t + \pi/4)$ Volts, and the 2Ω resistor is the load.

(a) [12 points] Find L and C that maximize power transfer to the load.

First, find the regular L-section



$$C' = \frac{1}{\omega_0^2 L (1 + \frac{1}{Q^2})} = \frac{1}{10^{14} \cdot 0.4 \times 10^{-6} \times 1.25} = \underline{\underline{20 \text{ nF}}}$$

$$10 = (1 + Q^2) \cdot 2 \Rightarrow Q = 2$$

$$Q = \frac{\omega_0 L}{R} = \frac{10^7 \cdot L}{2} = 2$$

$$\underline{\underline{L = 0.4 \mu\text{H}}}$$

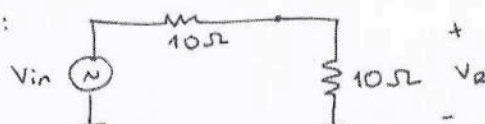
Finally, C' is parallel connection of C and 5 nF . So,

$$C' = C + 5 \text{ nF} \Rightarrow \underline{\underline{C = 15 \text{ nF}}}$$

(b) [5 points] What is the power delivered to the load for this maximum power transfer case?

$$P_d = P_A = \frac{|V_{in}|^2}{8R_s} = \frac{5^2}{8 \cdot 10} = 0.3125 \text{ W} = \underline{\underline{312.5 \text{ mW}}}$$

Alternatively:



This is what the source sees.

$$P_d = \frac{1}{2} \frac{|V_d|^2}{R} = \frac{1}{2} \frac{|V_{in}|^2}{R} = \frac{|V_{in}|^2}{8R} = 312.5 \text{ mW}$$

(c) [8 points] Now, instead of the modified L-section, we would like to implement a narrowband transformer for impedance matching between the source and the load. For transformer, we will use a core with $A_L = 5 \text{ nH/turns}^2$. Choose number of turns of the primary and secondary such that no additional capacitors or inductors are needed. You can leave the number of turns at the secondary as a non-integer number. Draw the entire circuit.

Use inductance of the primary (L_p) to tune out 5 nF capacitor.

$$L_p = \frac{1}{\omega_0^2 \cdot 5 \cdot 10^{-9}} = \frac{1}{10^{14} \cdot 5 \cdot 10^{-9}} = 2 \mu\text{H}$$

$$L_p = n_1^2 A_L \Rightarrow n_1 = \sqrt{\frac{2 \cdot 10^{-6}}{5 \cdot 10^{-9}}} = \underline{\underline{20}}$$

$$n_2 = \frac{20}{\sqrt{5}} = \underline{\underline{8.94}}$$

$$10 = \left(\frac{n_1}{n_2}\right)^2 \cdot 2 \Rightarrow \frac{n_1}{n_2} = \sqrt{\frac{10}{2}} = \sqrt{5}$$

